

Replication and Result Verification Guide

1. Data Source

The datasets used in this study are obtained directly from the laboratory repository (ideas-labo/ise-lab-solution). The provided notebook automatically downloads the raw .csv files via GitHub URLs. This ensures that the experiments start from standardized and unmodified source data, with a fixed random seed (random_state=999) used for shuffling to guarantee consistent data partitions during replication.

2. Experimental Stability

To reduce the impact of randomness, each model configuration (GaussianNB, MLP, and SVM) is evaluated over 30 independent runs. All reported metrics (Accuracy, Precision, Recall, Macro-F1, and AUC) are computed as the average across these runs, providing a stable and unbiased estimate of model performance.

3. Precomputed Results

Due to the significant computational cost of SBERT feature extraction and neural model training (the full execution of all six combinations can take nearly an hour, with the TensorFlow dataset typically requiring more time than PyTorch), the full experimental results are precomputed and stored in the results/ directory:

- **results/pytorch/:** Raw metric outputs for all model combinations on the PyTorch dataset.
- **results/tensorflow/:** Raw metric outputs for all model combinations on the TensorFlow dataset.
- **results/figures/:** Visualizations corresponding exactly to those presented in the main report.

4. Statistical Validation

Performance improvements are validated using the Wilcoxon signed-rank test. The implementation is included in the notebook, and the results support the statistical significance of the reported differences between feature sets (TF-IDF vs. SBERT).

5. Reproduction Procedure

To verify the integrity and reproducibility of the findings, the following levels of verification can be performed:

- **Level 1: Data Integrity Audit (Instant):**
Inspect the .csv files in the results/ directory. Calculate the mean values of the 30 runs and compare them with the tables provided in the main report.
- **Level 2: Visualization Verification (Fast)**
Follow "Option B" in manual.pdf. By uploading the precomputed CSV files to the Colab environment and running only the visualization cells, you can confirm that the raw data correctly generates the reported charts.

- **Level 3: Full Algorithmic Replication (Complete)**

Follow "Option A" in manual.pdf. Execute the entire pipeline to re-generate the features and re-train the models. While minor variances may occur due to hardware differences, the overall performance trends and statistical significance will remain consistent with the reported results.

6. Conclusion on Reproducibility

The combination of automated data fetching, repeated experimental loops, and the provision of raw result logs ensures that this triage tool meets high standards of computational reproducibility.